EV Market Performance & Insights

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EXECUTIVE TABLEAU ANALYTICS & STRATEGIC PROJECT PORTFOLIO

TOTAL GLOBAL SALES	AVERAGE VEHICLE PRICE	AVERAGE DRIVING RANGE	ANNUAL CO2 SAVED
1.558 B	$64,921	324.54 km	738 K Tons
▲Cumulative Units	Portfolio Mean MSRP	Certified Combined Cycle	▼Net Emissions Offset

1.

Executive Summary & Automotive Market Landscape

The global electric vehicle (EV) sector is transitioning into high-scale industrialization driven by carbon neutrality mandates, battery chemistry diversification, and surging consumer adoption. This comprehensive analytical project delivers end-to-end intelligence across vehicle segments, manufacturer distribution networks, and battery performance metrics. Analyzing billions of operational data points equips stakeholders with actionable intelligence to accelerate sustainable mobility deployments worldwide.

SEGMENT CONTRIBUTION: ANNUAL SALES UNITS BY VEHICLE CATEGORY — TABLEAU WORKSHEET: S1_CATEGORY_SALES



2. Multi-Stakeholder Intelligence Framework

Scenario 1 • Strategy

Emily Johnson

Focus: Segment portfolio optimization, driving range competitiveness, and customer satisfaction across high-growth passenger categories.

Scenario 2 • BI & Ops

David Chen

Focus: Manufacturer benchmark, manufacturing country distribution, market share velocity, and battery chemistry market dominance.

Scenario 3 • Sustainability

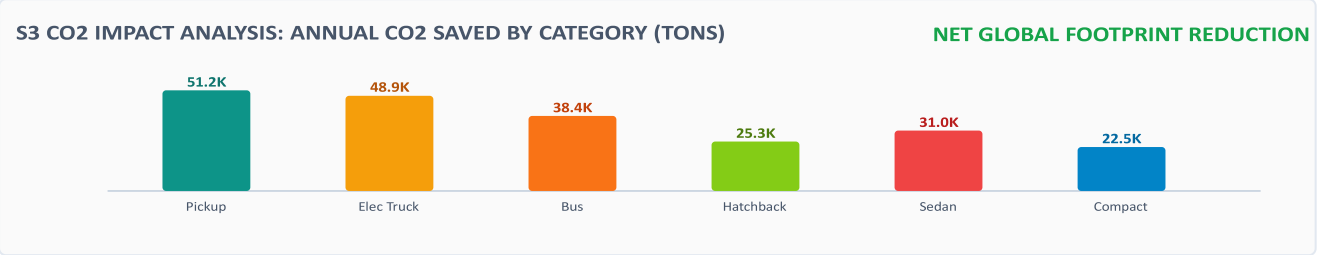
Dr. Sarah Williams

Focus: Lifecycle degradation, battery cycle durability, fast-charging energy trade-offs, and net territorial CO2 reduction.

3. Empirical Analysis: Battery Chemistries & Lifecycle Dynamics

Data synthesized from the Tableau dashboard reveals distinct trade-offs across battery chemical formulations. While **NMC (Nickel Manganese Cobalt)** and **LFP (Lithium Iron Phosphate)** dominate commercial vehicle sales, next-generation **SolidState** cells exhibit superior cycle endurance and thermal stability.

Battery Chemistry	Market Share Index	Avg. Cycle Life	Avg. Range (Sedan/SUV)	Degradation Rate
LFP (Lithium Iron Phos.)	High Volume Leader	2,950 Cycles	360 – 410 km	3.8% / 100k km
NMC (Nickel Manganese Cob.)	Premium Volume Leader	2,480 Cycles	440 – 510 km	5.1% / 100k km
NCA (Nickel Cobalt Alum.)	Specialized Performance	1,800 Cycles	460 – 520 km	7.4% / 100k km
Sodium-Ion	Emerging Entry Tier	1,600 Cycles	240 – 300 km	7.2% / 100k km
Solid-State	Next-Gen Validation	3,100 Cycles	520 – 610 km	2.2% / 100k km



4. Tableau Story Points & Strategic Findings

Story Point 1• Market Snapshot
Global sales hit 1.558B units at an average MSRP of $64,921. Production concentration remains led by key Asian and North American clusters.

Story Point 2• Value Retention
Sedans and SUVs dominate standard consumer range, whereas heavy commercial fleets (Electric Buses) secure highest 3-year retained value ($14.5M).

Story Point 3• Battery Performance
NMC and LFP dominate production volume. Solid-state technology guarantees the lowest degradation floor (<2.5% loss across 3,000 cycles).

Story Point 4• Environmental Impact
Fast charging architectures (~350 kW DC) expedite turnaround, while commercial conversions drive an annual global offset of 738,000 tons of CO2.

Tableau Public Storyboard Publication:
https://public.tableau.com/views/tab1_17930891578410/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link